

EXPOSÉ

Mapping Cross-Domain Solution Spaces: A Hierarchical Graph Approach to Biomimetic Analogy Discovery

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Abstract

Engineers facing complex design challenges often default to familiar domain-specific solutions while remaining unaware of optimized strategies evolved by nature or already solved in unrelated technological fields. Multiple barriers prevent this transfer at scale. Biologists and engineers describe functionally similar mechanisms in entirely different vocabularies, making keyword-based search ineffective. Beyond terminology, biology and engineering solve problems through fundamentally different strategies, so bridging the domains requires more than translation. Additionally, human memory retrieval is biased toward surface similarity, meaning that structurally relevant analogies from distant fields are precisely the ones least likely to be noticed.

This doctoral research aims at developing a hierarchical knowledge graph that maps the cross-domain solution space, connecting technological problems and solutions from patent literature with analogous biological strategies. We use large language models to extract problem and solution pairs from patent documents and generate semantic embeddings that capture functional meaning, allowing structurally similar solutions to cluster together regardless of their source domain. We then apply community detection on the graph to identify related solution clusters, which are generalized into higher-level abstractions through LLM-assisted summarization under human supervision.

By applying this process recursively, we construct a hierarchical graph that constitutes a taxonomy of the cross-domain problem-solution space, ranging from specific patent solutions to generalized problem-solving principles. Biological solutions extracted from scientific literature are integrated into this structure, enabling discovery of biomimetic analogies.

The resulting knowledge graph serves both as a tool for human exploration and as queryable infrastructure for agentic AI systems that can autonomously traverse the hierarchy, grounding analogical reasoning in verified data rather than parametric generation. The goal is to improve cross-domain knowledge transfer, with a focus on energy and resource efficient biomimetic solutions.

Keywords: analogy discovery, biomimetics, knowledge graph, patent analysis, bio-inspired design, cross-domain knowledge transfer, agentic AI

Research Context and Motivation

Engineering progress depends on finding good solutions to technical problems. Cross-domain knowledge transfer, drawing on solutions from fields outside one's own expertise, has been shown to improve both the quality and novelty of design outcomes (Fu et al., 2014; Hope et al., 2017). Biology is a particularly rich source of transferable solutions: Vincent et al. (2006) compared biological and technological strategies systematically using the TRIZ framework and found only 12% overlap in the principles used, meaning that biology offers a large reservoir of solution strategies that engineering has not yet adopted. Self-cleaning coatings inspired by the lotus leaf and adhesive technologies derived from gecko feet illustrate the practical value of such transfer.

However, several barriers prevent cross-domain transfer at scale. The most fundamental is that biology and engineering solve problems through different means. As the 12% overlap from Vincent et al. (2006) shows, the domains do not merely name solutions differently but rely on largely distinct sets of principles, so bridging them requires more than a dictionary. Second, biologists and engineers describe even functionally identical mechanisms in entirely different vocabularies (Chiu & Shu, 2007; Helms et al., 2009). A beetle's fog-collecting mechanism and an engineer's condensation harvester address the same problem but are invisible to each other through keyword-based search. Third, human memory retrieval is biased toward surface similarity, favoring within-domain matches over structurally similar but superficially different analogies from other fields (Fu et al., 2014). Far-field analogies, such as those between biology and engineering, are harder to retrieve, harder to notice as relevant, and harder to transfer successfully.

Knowledge graphs offer a promising approach to overcoming these barriers because they can represent structured relationships between problems, solutions, and domains in a way that supports both human exploration and computational querying. Chen et al. (2024) demonstrated that graph-based retrieval outperformed both the AskNature database (Biomimicry Institute, 2024) and semantic networks in supporting creative bio-inspired ideation, providing empirical evidence that structured knowledge representations can facilitate cross-domain design. More broadly, graph-based representations of functional relationships have been shown to improve creative ideation across large patent corpora (Sweed et al., 2025).

Recent work has begun to apply AI methods to construct and query such knowledge representations for design. Hope et al. (2017) showed that decomposing ideas into purpose and mechanism representations improves analogy retrieval compared to standard text similarity, and Bhavya et al. (2023) proposed LLM-based analogy generation and evaluation across technical domains. The AskNature database offers curated biological strategies organized by function but is limited in scale and relies on keyword-centric retrieval. The SEABIRD system (Vandevenne et al., 2015) automates biological analogy retrieval via NLP but lacks structured representation of the problem-solution space. Chen et al. (2024) built a knowledge graph for bio-inspired design using LLM-based entity extraction from AskNature and link prediction for

new biology-to-application mappings. Chen et al. (2025) extended this by training a fine-tuned LLM on graph-derived data to generate bio-inspired concepts directly. Sweed et al. (2025) constructed functional concept graphs over 500,000 patents, demonstrating that abstraction hierarchies within patent data support cross-domain ideation. In parallel, agentic AI systems that combine language models with planning, tool use, and iterative retrieval have shown the ability to query structured databases and ground their reasoning in verified data (Buehler, 2025; Singh et al., 2025), and Lenat & Marcus (2023) argue that trustworthy AI depends on combining flexible language models with structured knowledge representations.

Despite this progress, three specific limitations remain in the existing literature. First, no existing system constructs a scalable hierarchical abstraction of the problem-solution space that bridges biology and engineering. Current approaches either operate within a single domain (Hope et al., 2017; Sweed et al., 2025) or use flat ontologies with predefined entity types that cannot capture the multi-level structure of biological and technical solutions (Chen et al., 2024). Hope et al. (2017) explicitly note that their representation does not distinguish hierarchical levels of purpose and mechanism, which is particularly limiting for biological systems that are hierarchical from organelle to ecosystem level (Vincent et al., 2006). Second, existing systems lack either structured grounding or biological integration. The generative approach of Chen et al. (2025) produces bio-inspired concepts without traceable provenance, while patent-focused systems like that of Sweed et al. (2025) do not integrate biological solutions at all. Third, no agentic workflow exists that can leverage a cross-domain knowledge graph for grounded, multi-step analogical reasoning, where each proposed analogy is anchored in specific evidence from the graph rather than generated from statistical patterns alone.

This research addresses these three gaps by developing a hierarchical knowledge graph that maps the cross-domain solution space, connecting technological solutions from patent literature with analogous biological strategies. We use large language models to extract problem and solution pairs from patent documents and biological literature, generate semantic embeddings that capture functional meaning independent of domain-specific vocabulary, and apply recursive graph-based clustering with LLM-assisted generalization to construct a taxonomy from specific solutions to abstract problem-solving principles. The resulting graph serves both as a tool for human exploration and as queryable infrastructure for agentic AI systems that can traverse the hierarchy, decompose problems, and assemble cross-domain solutions grounded in verified data.

By bridging the vocabulary gap between biology and engineering at scale, this work aims to improve cross-domain knowledge transfer, with a focus on energy and resource efficient biomimetic solutions. The goal is not to replace human judgment but to surface potentially relevant connections from large text corpora that would otherwise remain hidden behind domain-specific terminology, enabling both human researchers and AI systems to discover biomimetic analogies based on functional similarity rather than keyword overlap.

Research Questions

This doctoral project develops and evaluates a hierarchical knowledge graph that connects technological solutions from patent literature with biological strategies. The work addresses four research questions.

RQ1: How can problems and solutions be extracted from heterogeneous text sources and represented in a unified semantic space?

Prior work on extracting and embedding problem-solution pairs has operated within single domains: Hope et al. (2017) worked with product descriptions, Bhavya et al. (2023) with technical concepts in cybersecurity and machine learning, and Sweed et al. (2025) with patent texts. None of these approaches has been applied to construct a unified representation that bridges patent and biological literature into a single queryable semantic space. The challenge is not only extraction and embedding, which are established techniques, but achieving a cross-domain representation where functional similarity between a biological adaptation and a technical invention corresponds to spatial proximity in the embedding space, despite fundamentally different source vocabularies. We assume that engineers describe their functional needs as free-text natural language problem statements, without requiring a predefined vocabulary or taxonomy. This means the cross-domain representation must be robust enough to bridge not only the distinct vocabularies of patent and biological literature but also the informal and varied language engineers use when formulating design problems. Structure Mapping Theory (Gentner, 1983) provides the theoretical foundation: analogies depend on shared relational structure rather than surface features, meaning a biological system and a technical device can be analogous when they solve similar problems through similar principles, even across domains.

RQ2: How can an interpretable taxonomy of the problem-solution space be constructed through graph-based clustering and LLM-assisted generalization?

Representing millions of individual patent solutions requires hierarchical organization. Related solutions should cluster together, and these clusters should be abstracted into higher-level problem-solving principles, yielding a taxonomy that classifies solutions by their functional role rather than their domain of origin. The challenge is constructing this hierarchy in a way that is both computationally scalable and semantically interpretable. The approach uses graph-based clustering and LLM-assisted summarization under human supervision, applied recursively to build abstractions from specific solutions to generalized problem-solving principles. A key question is how to determine the appropriate granularity at each hierarchical level, where appropriateness is defined by the resulting utility for analogy discovery by both human users (RQ3) and agentic systems (RQ4). This creates an iterative dependency: validation outcomes from RQ3 and RQ4 feed back into granularity and abstraction decisions, requiring multiple cycles of construction, evaluation, and refinement rather than a single pass.

RQ3: How can the resulting knowledge graph be validated for its utility in biomimetic analogy discovery?

The ultimate purpose of the knowledge graph is enabling discovery of biomimetic analogies. Validation must assess both the structural quality of the graph and its practical utility for this purpose. The methodology combines quantitative metrics with qualitative evaluation by domain experts from engineering, biology, and bio-inspired design, including a leave-one-out approach that removes known biomimetic innovations and tests whether the system can rediscover them. Challenges include defining appropriate success metrics and ensuring the validation captures real-world utility rather than artificial benchmarks.

RQ4: How can agentic AI systems leverage the hierarchical knowledge graph to perform grounded, multi-step biomimetic analogy discovery?

The knowledge graph from RQ1–RQ3 is designed not only for human exploration, but also as infrastructure for autonomous AI agents. An agentic system can decompose an engineering problem, navigate multiple abstraction levels, and retrieve specific biological strategies, with each proposed analogy traceable to concrete graph nodes. This addresses a core limitation of unconstrained language model generation: without structured grounding, outputs may sound plausible but lack verifiable evidence (Lenat & Marcus, 2023). The question is whether agent-driven discovery grounded in the graph produces more accurate and actionable biomimetic analogies than ungrounded generation alone.

The four research questions are interdependent and addressed through an iterative process rather than a strictly sequential pipeline. Validation results from RQ3 and RQ4 inform whether the hierarchical abstractions constructed in RQ2 support effective analogy discovery, and shortcomings feed back into adjustments of clustering granularity, abstraction strategies, and the underlying cross-domain representation from RQ1. This iterative cycle of construction, evaluation, and refinement continues until the knowledge graph demonstrably supports biomimetic analogy discovery for both human users and agentic systems.

Methods

The methodology combines computational text analysis with human validation. The project builds on a review of analogical reasoning theory, bio-inspired design research, and computational analogy mining, which informs evaluation criteria and guides approach selection. The technical work proceeds across four components.

Data Collection

The research requires two data sources: patent texts and biological solution descriptions.

For patents, we gather documents from USPTO (United States Patent Office) and EPO (European Patent Office). Because patent systems require publication, granted patents are publicly available under permissive licences that support large-scale scientific analysis. The actual size of the final dataset is to be determined based

on computational feasibility and theoretical saturation. After cleaning and preprocessing, we use large language models to extract the specific problems each patent addresses and the solutions it describes.

For biological solutions, the AskNature database (Biomimicry Institute, 2024) provides the most comprehensive listing of biological strategies, but automated access to its content is restricted, making it unsuitable for our approach. Additionally, the database lacks transparency and consistent scientific grounding. Therefore, we extract biological solution patterns from scientific literature by first identifying relevant scientific articles from openly available datasets, like Semantic Scholar, using topic modeling. Then we process selected papers through an adapted version of the bio-strategy-extractor from the NASA PeTaL project (Shyam et al., 2019).

Data preparation includes text preprocessing, extraction of problem and solution statements, embedding generation with instruction-tuned transformer models, and creation of evaluation datasets with known biomimetic innovations.

Hierarchical Graph Construction

The computational work proceeds iteratively across four stages.

First, we use large language models to identify the specific problems and solutions described in each patent document. This structured representation enables semantic analysis beyond surface-level text similarity.

Second, we project extracted problems and solutions into a common vector space using instruction-tuned transformer embedding models. These embeddings capture semantic meaning, allowing functionally similar solutions to cluster together regardless of domain-specific terminology.

Third, we use graph-based clustering to construct a network of problems, solutions, and their semantic distances. Community detection algorithms such as the Leiden algorithm identify clusters of related solutions. These clusters are validated against traditional clustering methods like HDBSCAN and k-means to ensure robustness.

Fourth, we infer common higher-level problems and solutions for each cluster through LLM-assisted generalization under human supervision. Machine learning supports the analysis, while final decisions remain with the research team. This approach handles large amounts of data while preserving scientific rigor. The generalized solutions are then re-embedded, allowing recursive clustering and generalization until the graph forms a taxonomy spanning levels from specific patent solutions to generalized problem-solving principles. Biological solutions extracted from scientific literature are integrated into this structure through their semantic embeddings, connecting them with technical solutions that are functionally similar.

Agentic Operationalization

Building on the knowledge graph, we develop a prototype agentic system that interfaces with the graph through standard query operations: semantic similarity search over embeddings, traversal of the hierarchical structure, and retrieval of node metadata including source patents and biological literature. Given a natural language description of an engineering problem or functional need, the system embeds the query into the same semantic space used for the graph’s problem-solution pairs. The agent then plans a retrieval strategy, queries the graph at multiple levels of abstraction, and synthesizes candidate biomimetic analogies. Each candidate is grounded in specific graph nodes, making the reasoning path transparent and auditable.

Evaluation

To ensure scientific rigor, we validate the proposed framework using a mixed-method approach combining quantitative metrics with qualitative expert validation.

Structural validation assesses the quality of the hierarchical graph. We validate cluster creation and generalization performance using a small subset of selected patents processed through our technique. The resulting graph network is analyzed by domain experts to identify shortcomings of the approach. Metrics include cluster coherence, coverage, and distinctiveness at each hierarchical level.

Utility validation tests whether the knowledge graph enables discovery of relevant biomimetic analogies. We evaluate the system’s capability using problems solved by known, historical biomimetic innovations that we explicitly remove from the graph beforehand. Success is measured by the system’s ability to propose solutions similar to the known biomimetic solutions. This leave-one-out approach provides a concrete benchmark for analogy discovery performance, applied to both human-driven and agent-driven exploration of the graph to enable comparison across discovery modes. For the agentic system specifically, we cross-validate using various configurations: agent with graph access, language model without graph access, and direct graph queries without agentic reasoning. This tests whether the combination of agentic reasoning and structured knowledge outperforms either component in isolation.

The validation process also examines reliability. Different runs of the clustering process and different evaluators should produce consistent results. Inconsistencies indicate either methodological problems or genuinely ambiguous boundaries in the solution space. Critically, evaluation is not a final step but part of an iterative cycle: validation outcomes from both human and agentic evaluation inform adjustments to clustering granularity and abstraction strategies, which are then re-evaluated until the graph demonstrably supports effective analogy discovery.

Work and Time Schedule

The project spans six semesters.

1st Semester: Literature review and theoretical framework development. Initial data collection establishes patent and biological literature corpora. Pilot experiments on extracting problem and solution statements, and on embedding methods, provide baselines and identify challenges. Technical infrastructure is established.

2nd Semester: Development and testing of graph clustering and LLM-assisted generalization methods. Multiple methods are compared and refined. Initial expert evaluation provides feedback on cluster quality and interpretability. First conference paper presents methodology and preliminary results.

3rd Semester: Scaling to larger patent corpora. Construction of full hierarchical graph through recursive generalization. First cycle of structural validation using metrics and expert assessment feeds back into granularity and abstraction adjustments. Development of the agentic prototype begins in parallel. First journal paper documents methodology and evaluation framework.

4th Semester: Integration of biological solutions into the knowledge graph. Leave-one-out evaluation using known biomimetic innovations, applied to both human-driven and agent-driven exploration. Validation results feed back into a second iteration of graph construction, refining hierarchical abstractions based on observed strengths and shortcomings.

5th Semester: Refined graph evaluation across discovery configurations (agentic, human, and baseline comparisons). Synthesis of findings across iterative cycles. Theoretical insights about computational support for cross-domain analogy discovery. Second journal paper presents complete system and findings.

6th Semester: Dissertation writing. Published papers form the core chapters. Introduction, discussion, and conclusion integrate findings and articulate contributions.

Expected Contributions

The resulting knowledge graph provides a taxonomy of the cross-domain problem-solution space and function-oriented retrieval that allows engineers to query by functional need rather than keyword matching, improving access to biological insights that are often obscured by domain-specific terminology.

The project combines large language models, semantic embeddings, and knowledge graph structure to support cross-domain knowledge transfer tasks. The agentic prototype further shows how structured knowledge representations can ground AI-assisted reasoning, so proposed analogies remain traceable to specific evidence rather than being generated from statistical patterns alone. This approach may

generalize beyond biomimetics to other domains where analogical reasoning across heterogeneous knowledge sources is valuable.

The methodology for constructing interpretable hierarchical abstractions from large text corpora through iterative human-AI collaboration contributes to the broader challenge of making AI-assisted analysis transparent and scientifically rigorous.

Declaration of Interest Statement

The author reports there are no competing interests to declare.

Supervision

Univ.-Prof. Dr.phil. Dr.rer.nat. Ulrich Frey from the Department of Environmental Systems Sciences at the University of Graz supervises this doctoral project.

Assoc. Prof. Dr.rer.nat. Georg Jäger from the Department of Environmental Systems Sciences at the University of Graz serves as second supervisor.

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